



DHARMSINH DESAI UNIVERSITY, NADIAD
FACULTY OF TECHNOLOGY
FIRST SESSIONAL
SUBJECT: (CE-701) Artificial Intelligence

Examination	: B.Tech Semester VII	Seat No.	:
Date	: 31/07/2023	Day	: Monday
Time	: 2:30 to 3:45 PM	Max. Marks	: 36

INSTRUCTIONS:

- Figures to the right indicate maximum marks for that question.
- The symbols used carry their usual meanings.
- Assume suitable data, if required & mention them clearly.
- Draw neat sketches wherever necessary.

- Q.1 Do as directed.**
- CO1 R (a) Define following properties of inference rules: i) Sound ii) Complete [12]
- CO1 U (b) Which generation language is PROLOG? Justify your answer. [2]
- (c) Assuming appropriate declarations are being made, state the output of the following PROLOG clauses for the goal a(X). [1]
- a(X) :- b(X), !, c(Y). [3]
- a(X) :- d(X).
- b(1).
- CO3 N b(4).
- c(1).
- c(X) :- fail.
- c(3).
- c(4).
- d(5).
- CO1 R (d) List the capabilities, which are required by machines to pass the Turing's test. Write the purpose of each capability listed. [2]
- CO1 R (e) Define following terms: i) Operationalization ii) Chronological backtracking [2]
- CO1 U (f) What are the arguments favoring use of heuristic? [2]
- Q.2 Write PROLOG programs for any 2 of the following:-** [12]
- CO4 C (a) Finding n^{th} element of a list. [6]
- Validate n, i.e. if it is not appropriate then a relevant message should be printed.
- Goal: nthelement ([1, 2, 3], 2) prints 2 as the output.
- CO4 C (b) Splitting list into two parts – 1st list contains only odd elements and second list contains only even elements. eg. oddeven ([1, 2, 3, 4, 5], X, Y) generates X = [1, 3, 5] Y = [2, 4], 1 solution as the output. [6]
- CO4 C (c) Rotating left elements of a list. eg. rl ([1, 2, 3, 4], X) generates X = [2, 3, 4, 1] as the output. [6]
- Q.3 Attempt the following questions.**
- CO2 A (a) Solve **BLACK+GREEN=ORANGE** crypt-arithmetic problem based on given rules: [12]
- Rules: Each letter or symbol represents only one digit throughout the problem. When letters are replaced by their digits, the resultant arithmetical operation must be correct. The numerical base is 10. Numbers must not begin with a zero. Show all necessary calculations. [6]
- CO2 N (b) i) Show how means ends analysis could be used to solve the problem of getting from one place to another. Assume that the available operators are walk, hire the auto, take a bus, take a train and take a flight. [3]
- CO2 N ii) Explain disadvantages of Hill Climbing and its solutions. [3]
- OR**

Q.3 Attempt the following questions.

CO2

A

- (a) Solve 8-puzzle problem using steepest ascent hill climbing algorithm. To calculate desirability of a move use following heuristic function.
 $H(X)$: Sum of Manhattan distance of the tiles from their goal positions.
 $H(\text{Start}) = h(1)+h(2)+h(3)+h(4)+h(5)+h(6)+h(7)+h(8)=0+1+1+0+1+1+0+1=5$
 $H(\text{Goal})=0$

Start State

1	5	2
4	8	3
7		6

→

Goal State

1	2	3
4	5	6
7	8	

Note: Tile's movement is restricted only to horizontal and/or vertical direction.
Show all necessary calculations.

CO2

N

(b)

i) Discuss following problem characteristic with respect to Travelling Salesman and Missionary and Cannibals problems.

1. Can solution steps be ignored or undone? Which class does it belong?
2. Is the universe predictable? Explain.
3. Is a good solution is absolute or relative? Explain.
4. Is the solution a State or a path? Explain.

CO2

N

ii) What are the issues in the design of search programs?



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FIRST SESSIONAL

SUBJECT: (CE - 714) Image Processing

Examination : B.Tech Semester VII
Date : 01-08-2023
Time : 02:30 PM To 03:45 PM

Seat No. :
Day : Tuesday
Max. Marks : 36

INSTRUCTIONS:

1. Figures to the right indicate maximum marks for that question.
2. The symbols used carry their usual meanings.
3. Assume suitable data, if required & mention them clearly.
4. Draw neat sketches wherever necessary.

Q.1 Do as directed.

- CO2 A (a) Validate that convolution in spatial domain is analogous to multiplication in frequency domain. [12]
- CO1 N (b) When you enter a dark theater on a bright day, it takes an appreciable interval of time before you can see well enough to find an empty seat. Which of the visual process is at play in this situation? [2]
- CO1 R (c) Consider an impulse function $\delta(t, z)$ of two continuous variables, t and z . Write the general form (equation) of $\delta(t, z)$ and sifting property of $\delta(t, z)$ at $(t, z) = 0$ and $(t, z) = (t_0, z_0)$. [2]
- CO1 U (d) Give the general form (equations) of Log and Power-Law intensity transformation functions and draw the graph representing these within the range $(0 \text{ to } L-1)$, where L is maximum intensity. [2]
- CO1 E (e) Consider the following 3 bit image of size 3×3 : [2]

2	7	3
3	4	7
1	2	4

Find the average value of intensities in the image using the probability information about the number of pixels being at a particular intensity level $(p(r_i))$.

- CO2 A (f) i. Is the filter shown below a smoothing filter or sharpening filter? Give brief reason. [2]

1	2	1
2	1	2
1	2	1

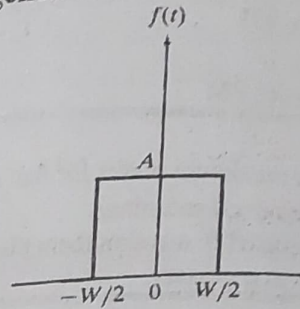
- ii. In a given application an averaging mask is applied to input images to reduce noise, and then a Laplacian mask is applied to enhance small details. Would the result be same if the order is reversed? [12]

Q.2 Attempt Any TWO from the following questions.

- CO1 C (a) The pixel intensity values of a gray level image have the probability density function $p_r(r)$ given by $p_r(r) = 2(1-r)$, for $0 \leq r \leq (L-1)$, and zero otherwise. It is desired to transform the gray levels of the image so that they have probability density function $p_z(z) = 2z$, for $0 \leq z \leq (L-1)$, and zero otherwise. Assume that r and z are continuous variables, [6]
1. Find the transformation that accomplishes this.
 2. Validate that $T(r)$ obtained after applying histogram equalization on $p_r(r)$ has uniform probability density function.
- CO1 E (b) Compute the lengths of the shortest 4-, 8- and m- path (if exists) between p and q [6]
- i. $V = \{0, 1\}$
 - ii. $V = \{1, 2\}$

3	1	2	1 (q)
2	2	0	2
1	2	1	1
(p) 1	0	1	2

CO1 E (c) Find the Fourier Transform of the box function in 1D shown in figure below and analyze the Fourier spectrum generated.



Q.3 Attempt the following.

CO2 E (a) The following figures show
fig 1: a 3-bit image of size 5 x 5
fig 2: a laplacian filter

	1	2	3	4	5
1	3	7	6	2	0
2	2	4	6	1	1
3	4	7	2	5	4
4	3	0	6	2	1
5	5	7	5	1	2

fig 1

Laplacian filter		
0	1	0
1	-4	1
0	1	0

fig 2

compute the following

- The output of a 3 x 3 mean (average) filter at (3,3)
- The output of a 3 x 3 laplacian filter shown in fig 2 at (3,3)
- The output of a 3 x 3 sobel mask at (3,3) i.e., G_x (Gradient in x direction)
- The output of a 3 x 3 sobel mask at (3,3) i.e., G_y (Gradient in y direction)
- Calculate Gradient magnitude.
- The output of median filter at (4, 5) with zero padding.

CO1 E (b) For the 8 x 8, 4 bit image shown below, draw the histogram. Show the new image along with its new histogram after histogram equalization.

Gray Level	0	1	2	3	4	5	6	7
No. of Pixels	9	8	11	4	10	15	4	3

OR

Q.3 Attempt the following.

CO2 U (a) Explain image formation model. Also discuss sampling and quantization to generate a digital image with suitable diagram.

CO1 E (b) a) Perform Histogram Equalization on the image with following data and draw the histogram of new image.

Gray Level	0	1	2	3	4	5	6	7
No. of Pixels	6	4	5	7	8	10	12	12

- b) For Histogram Equalization Transformation function $T(r)$ give reasons for the following conditions to be true.
- $T(r)$ is monotonically increasing in the interval $0 \leq r \leq L-1$.
 - $0 \leq T(r) \leq L-1$ for $0 \leq r \leq L-1$.



DHARMSINH DESAI UNIVERSITY, NADIAD
FACULTY OF TECHNOLOGY
FIRST SESSIONAL
SUBJECT: (CE725) BIG DATA ANALYTICS

Examination

Date

Time

: B.Tech Semester VII

: 02-08-2023

: 2:30 P.M. to 3:45 P.M.

Seat No.

Day

Max. Marks

:

: Wednesday

: 36

INSTRUCTIONS:

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Q.1 Do as directed.

[12]

CO1 U (a) Summarize the evolution of analytics/analysis: OLTP, OLAP ... RTAP. [2]

CO1 C (b) Describe how you will handle data migration from companyA to companyB of a given software product? [2]

CO1 A (c) Determine categories of Digital Data for the following via placing in the relevant basket of Structured, Unstructured or Semi-structured : Email, MS Access, Images, XML, Chat Conversations, HTML, Relations/Tables, Videos, MS Excel, JSON. [2]

CO5 R (d) How does HDFS handle datanode failure ? [2]

CO5 E (e) State True or False with justification: "Secondary namenode is a backup node." [2]

CO6 N (f) Differentiate the following. [2]
(I) Failed task vs killed task
(II) CopyFromLocal command vs MoveFromLocal command

Q.2 Attempt Any TWO from the following:

[12]

CO4 U (a) (I) Explain characteristics/V's of Big data in detail. (Any three). [3]
(II) Explain the similarities and differences between client server architecture and master slave architecture. [3]

CO5 A (b) Suppose a HDFS client wants to store the data of a file named as "demo.txt"(size=356MB) in HDFS, Assume that the system block size is configured as 128MB. Considering the above example illustrate the steps of HDFS Write mechanism in detail. [6]

CO6 C (c) Illustrate the steps of a MapReduce job that reads any text input and computes the **average length of all words that start with each character**. For any text input, the job should report the average length of words that begin with 'a', 'b', and so forth. case sensitive. [6]

For example ,

Mapper Input:-No now is definitely not the time

Reducer Output :-

N	2
d.	10
i	2
n	3
t	3.5

Q.3 Attempt the following:

CO2 A (a) Apply and explain in detail the Data Smoothing Technique using Equal Width Binning, Bin Means, and Bin Boundaries for the following Data:

5,10,11,13,15,35,50,55,72,92

CO3 E (b) Evaluate association rules on data items with Minimum Support threshold as 40% and Minimum Confidence level as 60% for the following transactions. Use Apriori algo for Market Basket Analysis.

Transaction ID	ITEMS List
T1	{Bread, Milk}
T2	{Bread, Donut, Book, Eggs}
T3	{Milk, Donut, Book, Cola}
T4	{Bread, Milk, Donut, Book}
T5	{Bread, Milk, Donut, Cola}

OR

Q.3 Attempt the following:

CO2 A (a) Apply the statistics of Five Number Summary (Upper Extreme, Lower Extreme, Median, Upper & Lower Quartile) for Interquartile range on the dataset {3, 7, 8, 5, 12, 14, 21, 13, 18}, and Visualize it using the Whisker/Box plot.

CO3 E (b) Support Data Characterization using Attribute oriented Induction (AOI) with example.



DHARMSINH DESAI UNIVERSITY, NADIAD

FACULTY OF TECHNOLOGY

B.TECH. SEMESTER VII [CE]

SUBJECT: (CE-718) COMPILER CONSTRUCTION

Examination : First Sessional

Date : 03/08/2023

Time : 02:30 PM to 03:45 PM

Seat No : _____

Day : Thursday

Max. Marks : 36

INSTRUCTIONS:

1. Figures to the right indicate maximum marks for that question.
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Q.1 Do as directed.

[12]

CO1-R a) Explain the following Terms with suitable example: Token, Lexeme and Pattern

2

CO1-A b) Identify the lexemes that make up the tokens from the given C code

2

int max (i, j)

{

return i > j ? i : j ;

}

CO1-U c) Compare: Compiler and Interpreter.

2

CO4-E d) State the number of nodes and edges in the DAG for the expression

2

$x + x + (x + x + x + (x + x + x + x))$

CO4-U e) Define strength reduction and frequency reduction with an example.

2

CO4-A f) Perform local copy propagation on the given code block with 4 instructions.

2

$p = q; r = p + q; s = p; p = r * s;$

Q.2 Attempt Any Two

[12]

CO1-R a) Explain phases of Compiler with a suitable block diagram.

6

CO1-A b) Construct Moore Machine for $f(x) = x \bmod 3$. Convert the Moore Machine to Mealy Machine.

6

Ex. $F(101) = 2$

$F(100) = 1$

$F(011) = 0$

CO1-C c) Construct Transition Diagrams for

6

(1) Relational Operators

(2) Identifiers

(3) White space

Q.3 Answer the following

[12]

CO4-A a) Translate the arithmetic expression $a = (-c * b) + (-c * d)$ into three address code, quadruple form and triple form.

6

CO4-E

b) Find local common subexpression elimination using AEB on the following code block.

(1)	$c = a + b$	(5)	$a = j + a$
(2)	$d = m \& n$	(6)	$j = b + d$
(3)	$e = b + d$	(7)	$a = -b$
(4)	$h = b + a$	(8)	If $m \& n$ goto L2

OR

Q.3 Answer the following

CO4-A

a) Compute EVAL(i) and KILL(i) for the graph in Fig. A.

CO4-E

b) Consider the source code in Fig. B. for dot product of two vectors a and b of length 20.

Do the following:

- Write the equivalent three address code
- Identify leader(s) and construct basic block(s)
- Draw the control flow graph

[12]

6

6

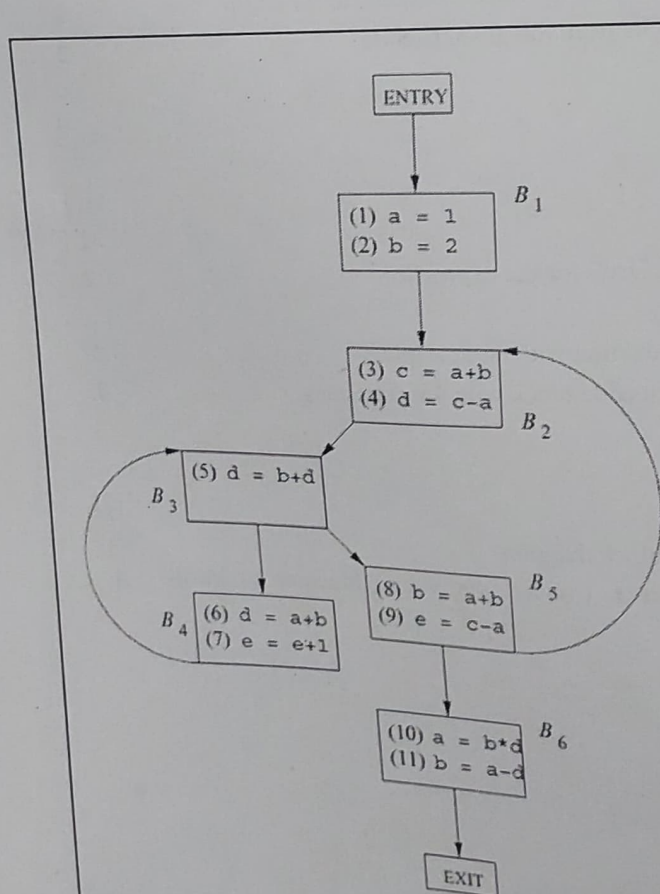


Fig. A.

```

begin
  prod := 0;
  i := 1;
  do begin
    prod := prod + a[i] * b[i];
    i := i + 1;
  end
  while i <= 20
end
  
```

Fig. B.